Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

This chapter presents the background and related literature of the study entitled “Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification.” It also presents the significance of the study, the statement of the problem, and the scope and limitations of the system.

1.1 Introduction

Property management involves maintaining tenant records, managing utility bills, computing charges, monitoring payment status, and providing billing information to tenants. These activities require the organized handling of property records, tenant information, billing documents, and other administrative records. Cammarata (2023) discusses the importance of records and information management in real-estate operations, while Teo et al. (2024) examine property-management functions and services. When property managers handle these activities through separate or manual methods, organizing information and maintaining consistent work processes can become more difficult.

To support these activities, digital approaches provide a way to centralize property information and organize regular management tasks. Property Technology (PropTech) refers to the use of digital technologies in real-estate and property-management

activities. Siniak et al. (2020) discuss the role of PropTech in changing real-estate operations and services, while Oyetunji et al. (2025) examine digital solutions for streamlining property-management operations in affordable-housing communities. Their discussion highlights how digital tools may support routine management activities, resource allocation, and tenant interaction, although the study also notes barriers such as financial constraints, infrastructure gaps, and resistance to change. Overall, this literature shows how digital technologies support property and housing-management activities and provides a basis for using centralized systems to organize property information and management processes.

One important area in property management is billing. Utility billing requires information from billing documents before charges can be processed, including amounts, dates, billing periods, providers, and account or reference identifiers. Optical Character Recognition (OCR) provides a way to extract text and relevant information from supported documents such as receipts and invoices (Kumar et al., 2020). Saout et al. (2024) discuss invoice-data extraction as a process involving document digitization, OCR, information extraction, and recognition of document structures. Mohan (2023) also presents a staged pipeline for extracting key-value information from commercial invoices. Overall, these studies support the use of OCR and document-processing methods to extract and organize information from billing documents for later processing.

After billing information is extracted, it still needs to be validated, classified, and processed according to defined billing conditions. Rule-based processing uses

predefined conditions and logical rules to support structured operational decisions and information handling. Baralis et al. (2022) discuss business-rule engines that represent and execute operational decisions in a structured way. In relation to process automation, Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation. These ideas are relevant to a billing workflow in which validation, classification, calculation, and exception-handling conditions are defined in advance. The reviewed literature provides a basis for combining OCR, rule-based processing, and notification technologies in a billing workflow for smaller residential property-management settings.

Based on these ideas, the proposed system connects property management, OCR-based extraction of supported utility-bill documents, validation, rule-based billing classification, billing computation, exception handling, and Web Push Notifications within one web-based workflow. In this study, the term “intelligent” refers to the combined use of OCR-enabled document text extraction and predefined, explicit, deterministic, and traceable rules for billing-workflow support. The OCR component may use machine-learning-based text-recognition technology to identify and extract available text from supported documents; however, OCR output is not treated as automatically validated billing data. Validation, duplicate checking, billing classification, billing computation, exception routing, and payment confirmation are handled through defined rules and Manager-controlled actions. The study focuses on organizing rental and utility billing activities within a centralized system while applying defined rules to billing and



exception-handling processes. Through this workflow, valid billing records can proceed automatically, while records requiring review are routed to the Exception Queue for Manager intervention.

1.2 Review of Related Literature

Digital Property Management and PropTech

Digital property management increasingly relies on technology to organize property details, management activities, billing data, and related services. Several studies have examined this shift from different perspectives: Teo et al. (2024) discuss technology and digitalization in property-management functions and service delivery, while Ding and Sun (2024) review technology innovations in housing management. In this context, PropTech refers to the use of digital technologies in the real estate and property-management sector. Siniak et al. (2020) examine the impact of PropTech on real estate industry growth, Saputra et al. (2023) review digital transformation trends in the property-management industry, and Al-haimi et al. (2025) synthesize the technologies, benefits, and challenges associated with digital transformation in real estate. In addition, Cammarata (2023) discusses the importance of records and information management in managing risks during real-estate changes. Overall, this literature supports the relevance of centralized digital tools for property information and administrative processes. Building on this literature, the present study proposes a centralized web-based workflow that organizes property, unit, tenant, utility, billing,

payment, exception, and receipt information for selected boarding houses and residential apartments.

OCR and Billing Document Extraction

Within property-management operations, billing is one area where digital processing can support routine administrative work. Billing documents contain information such as amounts, dates, billing periods, providers, and account or reference identifiers. Kumar et al. (2020) examine the extraction of information from bill receipts using Optical Character Recognition (OCR), while Saout et al. (2024) provide an overview of invoice-data extraction involving document digitization, OCR, information extraction, and document-structure recognition. Similarly, Mohan (2023) presents an end-to-end pipeline for custom key-value extraction from commercial invoices. However, extracting information from a billing document is only the first stage of billing processing. In the proposed system, OCR is used to process supported standard digital PDFs and clear images. After extraction, the information undergoes validation before billing processing. Records that do not meet the required validation conditions are sent to the Exception Queue for Manager review. Thus, OCR is treated as a method for extracting information from billing documents rather than as a guarantee that every extracted value is accurate. The OCR component may employ machine-learning-based text-recognition technology to convert visible characters and document text into machine-readable output. However, the proposed system does not use OCR output or machine-learning



recognition to determine billing validity, classify payment status, compute charges, or make property-management decisions.

Rule-Based Billing Processing

Once billing information has been extracted, it must be checked and processed according to defined conditions. Rule-based processing uses explicit conditions and decision rules to classify, validate, and route information. Wu et al. (2022) demonstrate the use of rule-based information extraction to organize domain-specific information, while Baralis et al. (2022) discuss business-rule engines for representing and executing operational decisions in a structured manner. Extending this idea to process automation, Geyer-Klingeberg et al. (2024) present a rule-based method that analyzes decision logic and classifies business processes according to their suitability for robotic process automation, while Kim et al. (2011) examine proactive exception handling in business processes. These studies support the use of predefined and traceable conditions to determine whether a record proceeds through normal processing or requires additional review. Applying these principles, the proposed system uses defined rules for billing validation, provider and account/reference mapping, duplicate detection, and identifying individual or shared billing arrangements. Records that do not meet the defined conditions are sent to the Exception Queue for Manager review. These rules, as well as the separation between Tenant Reported Paid and Manager-confirmed payment status, are system requirements rather than findings established by the cited studies. Digital receipts are created only after payment confirmation.

Shared Utility Billing and Cost Allocation

Shared utility billing presents a related issue because a single charge may need to be allocated among several occupants. Levinson and Niemann (2004) examine apartment energy use when landlords pay for utilities, while Mayr and Souza (2025) discuss coordination and incentive issues associated with shared heating costs. These studies provide context for understanding cost responsibility and consumption incentives in shared-utility arrangements. Nevertheless, they do not establish one universal allocation formula for rental properties or determine how the proposed system should calculate individual tenant charges. For this reason, the proposed system treats the allocation method as a defined system requirement rather than as a formula established by the reviewed literature. For shared occupancy, charges are split equally among eligible tenants in a Manager-configured group. The billing requirements also include reconciliation controls so that allocated charges equal the source billing amount. The allocation formula, rounding procedure, and reconciliation method are described in the succeeding chapters and will be tested during system evaluation.

Web Push Notifications and Secure Notification Delivery

After a billing record is finalized, tenants need to be informed that their billing information is available. The World Wide Web Consortium (2024a) describes the Push API, including push services and user subscriptions. The World Wide Web Consortium (2024b) also explains the role of service workers in handling web events in the background when they are included in the system implementation. Because Web Push

involves browser subscriptions and authorization, security must also be considered. Carboneri et al. (2023) examine Web Push implementations in the wild and identify security concerns involving push subscriptions and authorization. In the proposed system, Web Push Notifications serve as billing alerts that direct tenants to the authenticated web portal, where complete billing information is available. However, Web Push does not confirm payment, prove that payment occurred, or show whether the tenant read the notification. Tenant payment reporting therefore remains separate from Manager confirmation.

Iterative Software Development

Because the proposed system contains several connected functions, its development requires an organized and adaptable process. Iterative software development divides development into repeated cycles of building, reviewing, receiving feedback, and improving. Larman and Basili (2003) explain the development of iterative and incremental approaches, in which software is developed progressively through repeated cycles. Inayat et al. (2015) emphasize stakeholder involvement, communication, and requirements refinement in iterative requirements engineering, while Ramesh et al. (2021) identify challenges involving changing requirements, stakeholder participation, documentation, and communication in Agile software development. These studies support the use of Iterative Software Development Methodology in the present study. Although the literature supports this approach, the actual quality and performance

of the proposed system will be determined through implementation, testing, user feedback, and evaluation.

Software Quality Evaluation and Security

To evaluate the proposed system systematically, the study requires defined software-quality criteria. ISO/IEC 25010:2023 provides a product-quality model for systems and software quality requirements and evaluation (International Organization for Standardization, 2023). Guided by this model, the study will evaluate five quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. These characteristics provide a basis for examining whether the system performs its intended functions, performs efficiently, supports user interaction, operates consistently, and protects information. Security is particularly important because the system handles tenant and billing information. Republic Act No. 10173, or the Data Privacy Act of 2012, requires organizations that process personal information to use reasonable and appropriate measures to protect it (Republic of the Philippines, 2012). The National Privacy Commission (2016) further discusses safeguards for the confidentiality, integrity, and availability of personal data, while Zafar et al. (2022) review authorization and access-control strategies that restrict system functions and information according to user permissions. In the proposed system, role-based access separates Manager and Tenant functions. These sources guide the study's quality and security evaluation, but they do not prove that the system already meets the selected quality characteristics or privacy requirements. The Security evaluation is limited to the defined

system functions, access controls, and security-related criteria included in the study's evaluation procedures. It does not constitute a legal audit, certification, or claim of complete compliance with the Philippine Data Privacy Act.

1.3 Significance of the Study

The study develops and evaluates a web-based property-management workflow that uses OCR-based information extraction, rule-based billing classification, and Web Push Notifications. It primarily benefits property managers and tenants and may serve as a reference for future researchers and system developers.

In the global context, the study contributes an applied example of PropTech and digital property management by showing how existing technologies can be combined into one workflow for residential property operations. It also provides a local example of how digital property-management concepts can be applied in a smaller property-management setting. The study does not claim that the proposed workflow represents all property-management environments; rather, its significance lies in demonstrating how selected technologies may be integrated within the defined scope of the study.

In the economic context, the system is designed to reduce repetitive manual activities involved in receiving utility bills, recording billing information, calculating tenant charges, and communicating billing updates. Centralized records and predefined billing

rules can also help property managers follow a more consistent billing process. Processing time will be assessed during system evaluation. The study does not establish reductions in operating costs or late payments.

In the environmental context, the system uses electronic utility bills, digital billing records, and electronic notifications. This can reduce the need for paper-based records during the billing process. The study does not claim a specific reduction in paper use because this is not directly measured.

In the societal context, the system provides property managers and tenants with organized billing information. Itemized billing records, defined billing rules, and automated notifications can help make tenant dues easier to understand and communicate. This can support clearer billing transactions between property managers and tenants. However, the study does not claim that the system will eliminate billing misunderstandings or disputes because those outcomes are not directly measured.

1.4 Statement of the Problem

This study develops and evaluates an Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification for selected boarding houses and residential apartments in Pasig City.

Specifically, the study seeks to answer the following questions:

1. What development stages are undertaken in developing the proposed system using the Iterative Software Development Methodology?
2. How does the proposed system integrate OCR, rule-based billing processing, exception handling, payment management, digital receipts, and Web Push Notifications into one property-management workflow?
3. How does the developed system perform when evaluated using the selected ISO/IEC 25010:2023 quality characteristics in terms of:
    a. Functional Suitability;
    b. Performance Efficiency;
    c. Interaction Capability;
    d. Reliability; and
    e. Security?
4. What implementation guide and user manual can be prepared for the developed system?

1.5 Scope and Limitations of the Study

The study will develop and evaluate a web-based property-management workflow for selected boarding houses and residential apartments in Pasig City during Academic

Year 2026-2027. It will centralize property, unit, tenant, utility-mapping, rent, billing, payment, exception, correction, receipt, and notification records. “Intelligent” refers to OCR text extraction combined with explicit, deterministic, and traceable processing rules. An existing OCR engine may use machine-learning-based recognition, but the researchers will not train or customize a model. Validation, classification, computation, allocation and payment decisions will follow predefined rules and Manager-controlled actions rather than autonomous AI decisions.

Billing will cover rent, electricity and water only. Fixed, non-prorated rent will be generated once per monthly lease cycle on the lease-anniversary date, which will also be its due date. Short months will use their final calendar day while preserving the original anniversary day for later months. The first bill will show current rent and one-month advance rent separately. After the Manager confirms payment, the advance rent will be applied as credit to the final applicable rent cycle. No additional cash payment or duplicate receipt will be recorded. Individual charges will follow the configured lease or utility account; shared charges will use equal splitting among eligible tenants in a Manager-configured group. Shares will use two decimal places and must total the source charge.

Electronic utility bills will be retrieved from the designated property-management email account. OCR will extract the amount due, due date, billing period, provider and account or reference identifier from supported digitally generated PDFs and clear images. Required-field, mapping, period and duplicate checks will precede classification



and computation. Incomplete, uncertain, duplicated, unmappable or otherwise invalid records will enter the Exception Queue for Manager correction, revalidation and reprocessing. Finalized corrections will preserve prior values and payment history.

The bill payment statuses will remain Unpaid, Tenant Reported Paid, and Paid. Only the Manager may confirm a reported payment and authorize its receipt after any required verification outside the system. Rejected reports will return to Unpaid with a recorded reason. The system will not independently verify payments. Tenants will access their billing details, payment status, and confirmed receipts through the authenticated portal. Web Push will provide billing updates and reminders three calendar days before the due date for eligible unpaid records available before the scheduled reminder time. Rent generated on its due date will receive a billing update. Push permission, device support, or delivery failure will not prevent portal access, and an alert will not prove payment or readership. Tenant SMS and email notifications are excluded.

Admin access will be limited to system maintenance. The Manager will manage property and tenant records, mappings, billing rules, exceptions, corrections, payment confirmation and receipts. Tenants will view their permitted records, report payments, and receive permitted alerts. Authentication and role restrictions will protect the centralized records. Evaluation will cover Functional Suitability, Performance Efficiency, Interaction Capability, Reliability and Security under the selected ISO/IEC 25010:2023 characteristics. It will not establish long-term changes in revenue, profitability, payment compliance or tenant behavior.

The study excludes mobile applications; installation or physical reading of smart meters or submeters; Internet or Wi-Fi billing; predictive profit analytics or billing; Generative AI and autonomous AI billing decisions; blockchain; automatic utility disconnection; payment gateways or bank integration; independent external payment verification; maintenance tickets; legal contract generation; and automated eviction. OCR will not cover handwritten, heavily damaged, severely distorted, or low-resolution documents. The study does not guarantee notification delivery or readership, establish a universally fair utility-allocation formula, or provide complete legal certification under the Philippine Data Privacy Act.

CHAPTER II

METHODOLOGY

This chapter presents the methodology and procedures that will guide the development of the proposed Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. It describes the project requirements, system design, conceptual operation, interface mockups, and Iterative Software Development Methodology. It also presents the testing and evaluation procedures that will be used to examine the developed system within the approved scope of the study.

2.0 Project Overview

Project Information

The project, entitled Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification, is a web-based system for selected boarding houses and residential apartments in Pasig City. It supports property, unit, tenant, rent, electricity, water, utility-mapping, billing, payment-status, exception, receipt, and applicable notification records. The system provides an integrated workflow for rent and utility billing. It receives electronic utility bills through the designated property-management email account, retrieves supported attachments,

extracts available billing information through OCR, validates and processes records using predefined rules, computes and finalizes billing records, manages exceptions, records payment status, generates receipts after Manager-confirmed payment, and provides permitted Web Push billing and due-date alerts.

In this study, intelligent refers to OCR-enabled document text extraction combined with predefined, explicit, deterministic, and traceable rules. The OCR component may use machine-learning-based text recognition to extract available document information, but it does not make autonomous billing, validation, allocation, payment, or property-management decisions. These functions use defined rules and Manager-controlled actions.

Requirements Definition

The proposed system is designed to meet the following business, user, and security requirements.

A. Business Requirements

The system shall retrieve supported utility-bill attachments from the designated property-management mailbox and use OCR to extract the amount due, due date, billing period, provider and account or reference identifier. Explicit, deterministic and traceable rules shall check required values, provider/account mappings, periods and duplicates, identify the billing category and individual or shared arrangement, and route unprocessable records to the Exception Queue. Only valid records shall proceed to



computation and finalization. Manager corrections shall undergo revalidation and, where applicable, reprocessing or recomputation, including corrections to finalized records.

Billing shall cover rent, electricity and water. Fixed, non-prorated rent shall be generated once per lease cycle on the anniversary date, using the last calendar day when necessary and retaining the original day for later months. One-month advance rent shall appear separately on the first bill and become final-cycle credit after payment confirmation. Shared charges shall be split equally among eligible tenants in the configured group, using the rent-generation date for rent and the final day of the source period for utilities. The eligible list shall be stored. Computation in whole centavos and remainder distribution in ascending tenant-record identifier order shall preserve the source total. Missing groups or eligible tenants shall require exception review. Duplicate billing and repeated use of advance credit shall be prevented.

The centralized database shall retain property, unit, tenant, mapping, rent, billing, payment-status, exception, correction, receipt and applicable notification records. Bills shall use Unpaid, Tenant Reported Paid and Paid, with reported payments remaining unconfirmed until Manager confirmation. Receipts shall be issued only for confirmed payments. Tenants shall access billing details, statuses and confirmed receipts through the authenticated portal. Permitted Web Push billing updates and reminders three calendar days before the due date shall direct them to that portal. Routine valid records shall proceed automatically, while exceptions require Manager intervention. SMS, GSM and tenant email notifications shall not be used; the mailbox shall serve utility intake only.



B. User Requirements

The Admin shall perform only the implemented system-maintenance functions. The Manager shall log in with valid credentials and manage properties, units, tenants, utility providers, account mappings, rent, billing arrangements and applicable rules. The Manager shall review billing and OCR results, resolve exceptions, correct billing or payment information, submit corrections for revalidation and recomputation where applicable, confirm tenant-reported payments, and view payment statuses and confirmed receipts.

The Tenant shall log in with valid credentials and view assigned property or unit information, rent and utility bills, categories, periods, amounts due and payment statuses. The Tenant shall report an applicable payment, view confirmed-payment receipts, and receive permitted Web Push billing and due-date alerts linking to the authenticated portal. Reporting shall change a bill to Tenant Reported Paid rather than Paid. Manager confirmation shall mark it Paid, while rejection with a recorded reason shall return it to Unpaid.

C. Security Requirements

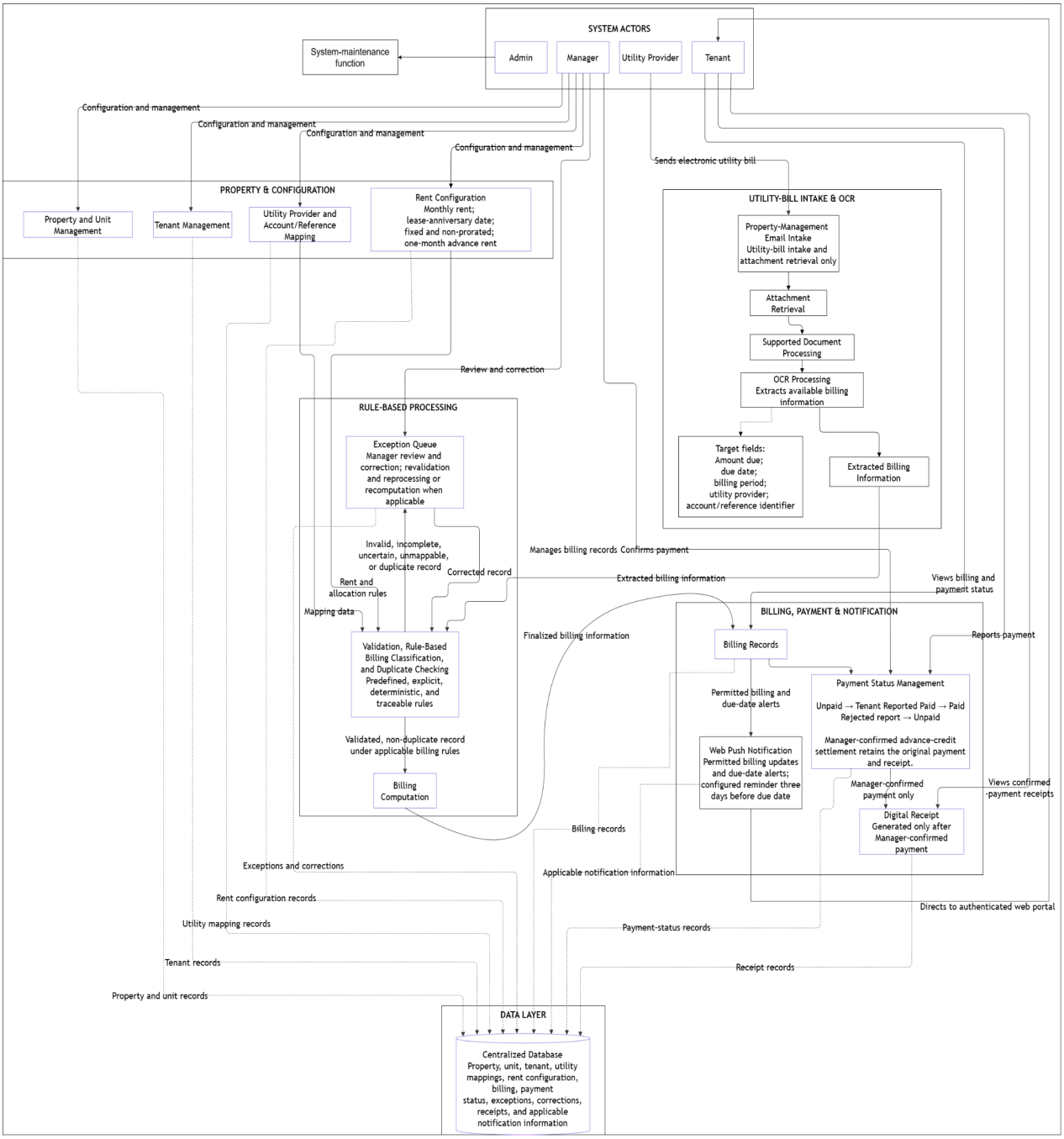
The system shall require valid credentials, restrict functions and records according to the Admin, Manager and Tenant roles, and limit property, tenant, utility, billing, payment, exception, correction, receipt, notification and source-document access to authorized users and functions. Authentication, defined permissions and session management shall protect records from unauthorized access or modification.

User-provided and extracted information shall be validated before storage or processing; failed field, identification, mapping, duplicate or other required checks shall route records to the Exception Queue. Correction information shall be retained in the centralized database. These requirements cover the implemented controls and evaluation criteria and do not constitute legal certification or a complete privacy audit.

2.1 Concept of the Study

Architectural Framework

The framework connects property and unit management, tenant management, utility-provider and account mapping, and rent configuration with email intake, OCR, validation, rule-based classification, duplicate checking, billing computation and the Exception Queue. Billing records, payment-status management, digital receipts and Web Push notifications use the Centralized Database. The Admin, Manager, Tenant and Utility Provider interact with these components according to their defined responsibilities.



Property and Unit Management manages property and unit information. Tenant Management manages tenant information and its association with the applicable property or unit. The Utility Provider and Account/Reference Mapping component



manages utility providers, account or reference identifiers, and their configured property or unit relationships. The Manager configures the applicable mappings.

Rent will be generated once per monthly lease cycle on the configured anniversary date, which will also be its due date. The charge will be fixed and non-prorated, with its coverage period recorded. If the anniversary day is absent, the system will use the month's last day and retain the original day for later months; a January 31 anniversary therefore uses February 28 or 29 and then March 31. Generation checks will prevent duplicate cycle charges, repeated advance-rent lines and new cycles after the lease end date. Existing unpaid charges will remain accessible, and membership changes will not silently replace finalized bills.

The first bill will show current rent and one-month advance rent as separate rent lines. Advance rent is not a security deposit and will become final-cycle credit only after payment confirmation. Shared tenants will have separate current and advance shares. When the Manager records the final cycle, the system will display its rent, available confirmed credit and remaining balance. Manager-confirmed credit application that fully covers the charge will mark it Paid using the original payment and receipt reference. It will neither record another cash payment nor create another receipt. Credit cannot exceed the available confirmed amount; only a remaining balance may be reported for payment. Discrepancies will follow Manager correction.

The designated property-management mailbox will retrieve supported utility-bill attachments and will not send tenant notifications. An existing OCR engine will recognize

text from selected provider layouts in digitally generated PDFs and clear images, without researcher training or customization of a machine-learning model. Predefined extraction rules will obtain the amount due, due date, billing period, provider and account or reference identifier, retaining the source reference. OCR will not establish payment status, billing validity or tenant charges; extracted fields will undergo validation and mapping.

Absent, unreadable or conflicting values, and unsupported attachments, will require exception review rather than finalized charges. Supported layouts, file types and processing limits will be documented before final testing. Evaluation will compare extracted fields with manually checked samples and report results only within the declared document boundaries.

Validation will check required fields, currency amounts, valid dates and a billing period whose beginning does not follow its end. Account identifiers will remain text to preserve leading zeros. A unique Manager-configured provider/account mapping will identify electricity or water and the covered property or unit; its saved configuration will determine the billing arrangement. Provider aliases will follow an explicit list. Missing, conflicting, invalid or unmappable values will enter the Exception Queue rather than being guessed. Rent will follow saved configuration without OCR classification.

Duplicate checks will compare attachments and provider/account/period combinations with existing records. Repeated or revised bills will require Manager review before another charge or correction is accepted, and reprocessing an accepted source

will not create duplicate allocations. Each result will retain its source reference, normalized fields and applied rule identifiers. Corrected records must pass the same checks before computation and finalization.

Table 2.1: Validation and classification rules

Rule ID	Rule Name	Description	Action/Result
V01	Required information	Amount, due date, billing period, provider, or account identifier is missing or conflicting	Exception Queue; no finalization
V02	Valid values	Amount or date cannot be interpreted, amount is negative, or period begins after it ends	Exception Queue; Manager review
V03	Unique mapping	Provider/account identifies exactly one configured target and category	Continue; otherwise Exception Queue
V04	Duplicate check	Identical attachment or same provider/account/period is already recorded	Hold for duplicate or revised-bill review
C01	Utility category	Configured provider/account mapping identifies electricity or water	Assign mapped category

Rule ID	Rule Name	Description	Action/Result
C02	Billing arrangement	Target has a valid individual/shared configuration	Apply saved arrangement; otherwise Exception Queue
A01	Allocation	Eligible group and reconciled shares are available	Proceed to finalization; otherwise Exception Queue
R01	Rent cycle	Lease/cycle already has a rent charge	Do not generate a second charge

Individual charges will follow the configured lease or utility account. Shared rent and utility charges will be split equally among eligible tenants in the Manager-configured property or unit group, without consumption estimates or meter readings. Eligibility will use active assignments on the rent-generation date for rent and on the source period's final day for utilities, excluding assignments already ended. The system will retain the eligible list with the allocation so later occupancy changes do not alter finalized charges. Missing groups, periods or eligible tenants will enter the Exception Queue.

Computation will divide the source charge in whole centavos, assigning equal base shares and distributing leftover centavos in ascending permanent tenant-record identifier order. PHP 100.00 split among three tenants will produce PHP 33.34, PHP

33.33 and PHP 33.33. Shares will display two decimal places and must total the source charge before finalization.

The Exception Queue will block finalization until Manager review and successful validation. Corrections will retain the affected field, previous and new values, reason and Manager identity, then undergo revalidation and applicable reprocessing or recomputation. Unresolved records will remain blocked with their failure reason; submitting a correction alone will not establish validity.

Billing Records will retain current and finalized charges using Unpaid, Tenant Reported Paid and Paid. Tenants may report full payment of a selected outstanding balance; partial and overpayments will not be processed automatically. Reports will remain unconfirmed pending Manager verification outside the system. Rejection will record a reason, restore Unpaid and preserve report history. Manager confirmation will mark the bill Paid and retain the confirmed payment amount, date, Manager identity, and unique receipt. Repeated confirmation will return the existing receipt. The system will not independently verify cash, bank transfers, electronic-wallet transactions or utility payments.

Finalized corrections will preserve prior values and processing history. Unpaid bills may be corrected after revalidation and recomputation; pending reports must be reviewed against any revised amount. Paid bills will retain their original payment and receipt, with corrections recorded as linked revisions showing the charge and balance

changes. External payments or refunds will remain Manager-handled without automated settlement. Tenants will access current records through the authenticated portal.

The Centralized Database will relate properties, units, tenants, leases, utility mappings, billing sources, allocations, payment reports, confirmations, receipts, exceptions, corrections and notifications. Recorded assignments will support multiple tenants per shared unit. Allocations will link to their source and eligible group; payments and receipts will link to their bills. These relationships will preserve transaction history and support access restrictions. Corrections will remain centralized records rather than a separate architectural component. The database design will define storage relationships while the framework, flowchart and DFD describe structure and operation.

Web Push will be the only external tenant-notification channel. On a supported browser, tenant permission will allow creation and storage of an account/device subscription and the background handling needed to receive an alert. Alerts will open the authenticated billing record without revealing complete details beforehand or confirming payment. Finalization and relevant corrections will trigger billing updates. Philippine-time reminders will be scheduled three calendar days before due dates for outstanding Unpaid records, excluding reports awaiting confirmation. Corrected dates will replace pending schedules, and repeated events will not duplicate reminders. Bills available after the reminder time, including rent generated on its due date, will receive billing updates rather than backdated reminders.

Notification records will distinguish scheduling, attempts and available push-service acceptance or failure results. Acceptance will not prove receipt or reading. Invalid subscriptions, denied permission, unsupported devices and delivery failures will not prevent authenticated access to bills, payment statuses or confirmed receipts. SMS, GSM and tenant email will remain excluded.

The Admin performs system-level maintenance. The Manager manages property information, tenant records, utility mappings, billing configurations, billing records, exceptions, corrections, payment confirmations, and receipts. The Tenant views billing information and payment status, reports applicable payments, views confirmed-payment receipts, and receives permitted Web Push Notifications. The Utility Provider serves as the source of electronic utility-bill documents received through the designated property-management email account.

The components operate as an integrated web-based property-management system. Property and tenant management, utility mapping, and rent configuration provide the required management information. Billing-processing components support utility-bill intake, OCR extraction, validation, rule-based classification, duplicate checking, billing computation, and exception handling. Billing and payment components manage billing records, payment statuses, and confirmed-payment receipts. The Web Push Notification component supports permitted billing and due-date-related alerts, while the Centralized Database stores the applicable system records. The Admin, Manager, Tenant, and Utility Provider interact with the system according to their defined roles and responsibilities.

2.2 Conceptual Operation

The workflow uses property, tenant, utility-mapping and rent configuration to process rent, electricity and water charges. Monthly rent follows the fixed anniversary rule, with advance rent shown separately on the first bill. Utility processing retrieves supported email attachments, extracts their fields through OCR, and validates required information, mappings, periods and duplicates. Valid records proceed to classification, computation and finalization; exceptions require Manager correction and renewed validation.

Shared charges use equal splitting among configured eligible tenants, with eligibility determined on the rent-generation date or the utility period's final day. Whole-centavo computation and consistent remainder distribution ensure that displayed shares total the source amount. Finalized charges begin as Unpaid. When a tenant reports payment, the status changes to Tenant Reported Paid. Only Manager confirmation changes the status to Paid and authorizes a receipt. Rejected reports return to Unpaid with retained history. The final rent covered by the confirmed advance credit follows the manager-confirmed credit application using the original payment and receipt, without another cash transaction or duplicate receipt. Permitted billing alerts and eligible reminders three calendar days before the due date link to the authenticated portal without proof of payment, delivery, or reading.

Process Flowchart

The process flowchart presents the order of activities and decisions in the billing workflow. It shows the separate rent and utility-bill paths, their convergence at Billing Finalization, the Exception Queue return to validation, payment-status transitions, Digital Receipt generation, and the supporting Web Push Notification branch.

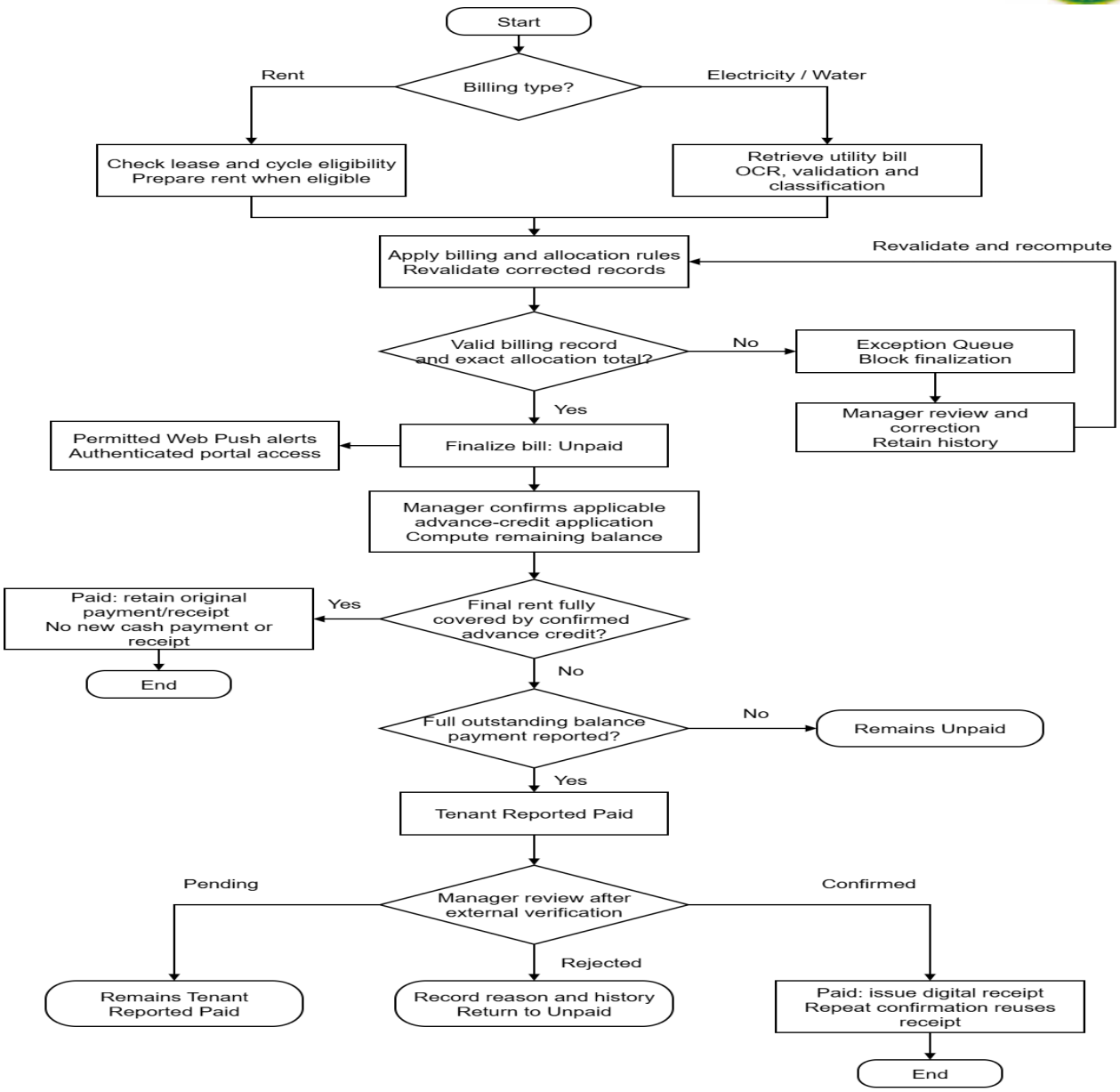


Figure 2.2. Overall Process Flowchart

The flowchart shows the sequence of billing activities and decision points from rent or utility-bill processing through Billing Finalization. It also shows that only Manager

confirmation changes a tenant-reported payment to Paid and permits Digital Receipt generation, while Web Push Notification remains a supporting alert function.

Full-Blown Data Flow Diagram

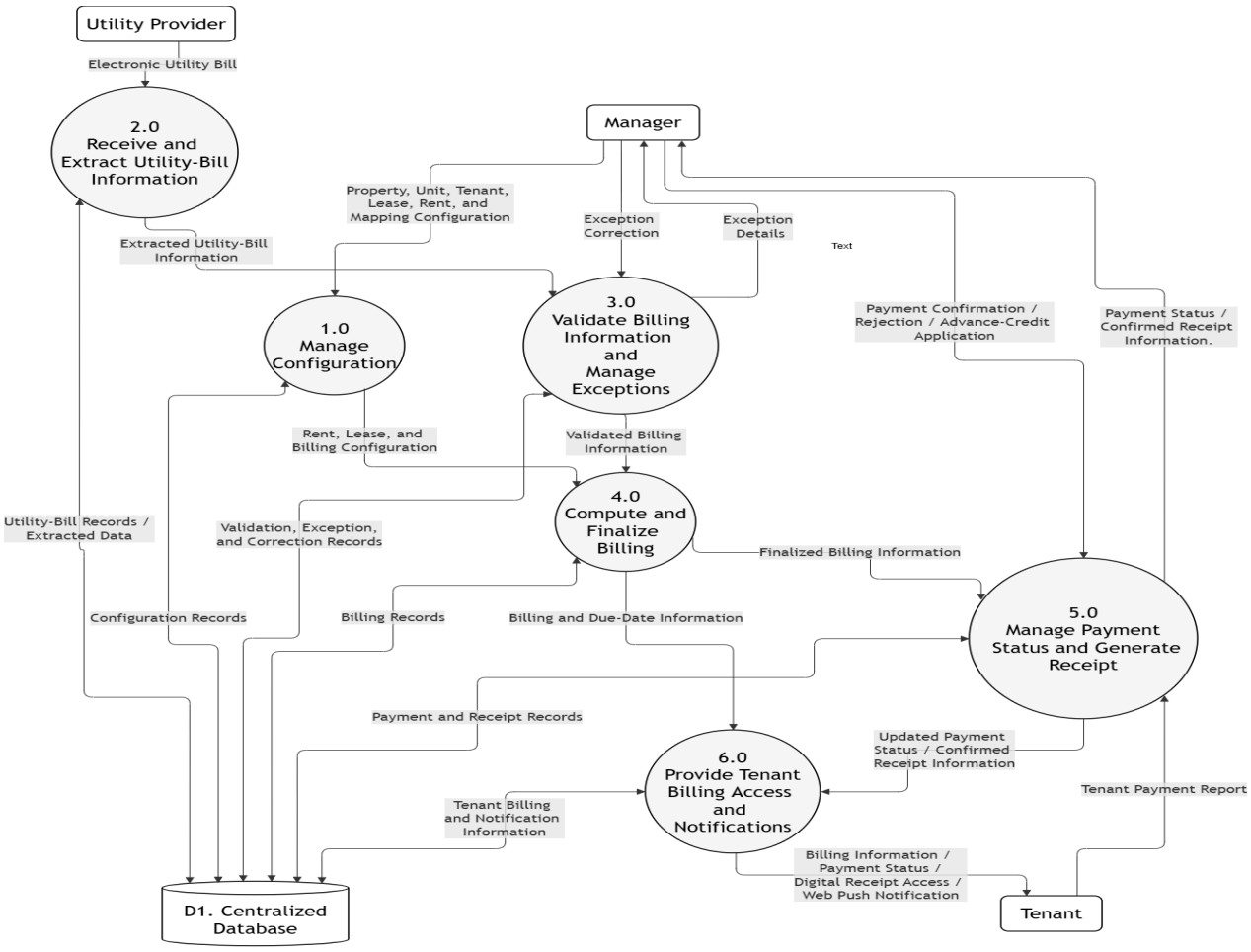


Figure 2.3. Data Flow Diagram

The DFD illustrates how billing-related information moves among the Manager, Tenant, Utility Provider, system processes, and Centralized Database. It shows the flow of configuration data, utility-bill information, billing records, payment information, receipt

information, and tenant notification information through the system processes. External entities interact with the system through the appropriate processes, while the Centralized Database stores and provides the records required by the system.

2.3 Mockup Design

The mockups present the proposed interfaces for the Admin, Manager, and Tenant roles. The Admin interface is limited to the system-maintenance functions included in the study. The Manager interface presents property and tenant records, billing configuration, incoming utility-bill information, the Exception Queue, billing records, and payment confirmation. The Tenant interface presents the assigned billing records, payment-reporting function, confirmed-payment receipts, and applicable notifications. The mockups serve as interface references during development and do not represent completed or evaluated system functions. Their labels, available actions, and displayed payment statuses will follow the approved scope and workflow.

2.4 Methodology of the Study

The study will use the Iterative Software Development Methodology to develop the proposed Intelligent Property Management Workflow with OCR-Enabled Automated Billing Classification and Tenant Notification. This methodology will organize

development into successive iterations in which related system functions will be planned, designed, developed, reviewed, refined, and integrated.

The methodology is appropriate for the proposed system because its functions are interconnected but can be developed in manageable groups. These functions include system foundation, property and billing configuration, utility-bill intake, OCR extraction, rule-based processing, exception handling, Billing Computation, payment and receipt management, tenant access, Centralized Database functions, and Web Push Notifications. The methodology will also support the intended automate by default, intervene by exception approach.

2.4.1 Principles of the Iterative Methodology

The Iterative Software Development Methodology will be guided by four development principles and the system's exception-based workflow. First, incremental development will allow the researchers to develop related system functions in smaller iterations instead of implementing the entire system at once. Second, continuous testing and review will allow each iteration to be checked against the defined requirements, user roles, workflow rules, and data requirements.

Third, refinement between iterations will allow issues identified during development to be corrected before related functions are integrated into succeeding iterations. Fourth, cumulative integration will allow functions developed in earlier iterations to work with functions developed later until they form a complete system. Throughout the iterations, development and testing will preserve the system's

exception-based workflow, allowing valid records to proceed while records requiring review enter the Exception Queue for Manager intervention.

2.4.2 Development Iterations of the Methodology

The proposed system will be developed through seven planned iterations based on the functional requirements of the study. Each iteration will focus on a related group of functions that will later be integrated into the complete system.

Table 2.2: Development Iterations of the Proposed System

Iteration	**Development Focus**	**Main Functions**
Iteration 1	Requirements and System Foundation	User roles, authentication, property, unit, tenant, and Centralized Database structure
Iteration 2	Property and Billing Configuration	Utility providers, account or reference mapping, rent configuration, billing arrangements, and billing rules
Iteration 3	Property-Management Email Intake and OCR	Designated email intake, attachment retrieval, supported document processing, and OCR extraction
Iteration 4	Rule-Based Validation and Exception Handling	Required-field checking, mapping, billing-period checking, duplicate checking, billing classification, Exception Queue, Manager correction, and revalidation
Iteration 5	Billing Computation and Finalization	Rent, electricity, and water billing; Individual and Shared-Occupancy Billing Arrangements; allocation; computation; and Billing Finalization

Iteration	Development Focus	Main Functions
Iteration 6	Payment, Digital Receipt, and Web Push Notification	Payment statuses, Tenant payment reporting, Manager confirmation, Digital Receipt generation, tenant portal access, and Web Push alerts
Iteration 7	System Integration and Testing	Integration of system functions, implemented security controls, system-wide testing, and final evaluation

2.4.3 Application of the Methodology to the Study

The planned iterations will be applied progressively, beginning with the system foundation and configuration functions and continuing with utility-bill intake, OCR extraction, rule-based processing, exception handling, Billing Computation, payment and receipt management, tenant access, and Web Push Notifications. Each iteration will build upon related functions developed in earlier iterations.

During development, each iteration will undergo planning, design, development, testing and review, followed by refinement and integration. Findings from testing and review will be used to correct related functions before succeeding components are integrated. Records requiring Manager intervention will be handled through the Exception Queue according to the defined processing rules.

The final iteration will integrate the developed functions and conduct system-wide testing and evaluation. The completed system will be evaluated using the selected

ISO/IEC 25010:2023 quality characteristics: Functional Suitability, Performance Efficiency, Interaction Capability, Reliability, and Security. No completed iteration, testing result, or evaluation result is claimed in this chapter.

2.4.4 Benefits of the Methodology to the Study

The Iterative Software Development Methodology will allow the researchers to develop the proposed system in manageable groups of related functions. This will support the early identification and correction of defects and inconsistencies before succeeding functions are integrated.

The methodology will also support continuous refinement as development progresses. Testing and review during each iteration will provide opportunities to improve related functions and maintain consistency among the system requirements, workflows, data structures, and user roles.

Finally, the methodology is appropriate for the proposed system because its functions are interconnected and require gradual integration. The iterative approach will allow the researchers to progressively form the complete system while maintaining the intended automate-by-default, intervene-by-exception approach.

2.4.5 System Testing and Evaluation Procedures

Testing will occur during each iteration and after integration, using business, user and security requirements. Cases will cover normal transactions, incomplete fields, unmapped accounts, duplicates and corrected exceptions, recording inputs, expected and observed results, and pass or failure. Defects will be corrected and affected tests repeated before final evaluation. OCR results for the five target fields will be compared with manually checked documents. Field accuracy will equal correctly extracted values divided by expected values, multiplied by one hundred; missing and incorrect values will be recorded. Declared normalization rules must preserve meaning. Document sets and rules will be fixed before final testing, with refinement samples identified separately from final evaluation samples.

Validation, mapping, classification and exception routing will be checked against predefined outcomes. Rent and utility computations will be checked independently, including individual shares and exact source reconciliation. Payment tests will verify Manager-only confirmation, correct advance-credit settlement and receipts only for confirmed payments. Repeated processing will be checked for duplicate charges or receipts.

Functional Suitability will use requirement tests; Performance Efficiency will use processing and response measurements under a stated environment; Interaction Capability will use representative tasks and an adviser-reviewed questionnaire; Reliability will use repeated-processing and recovery tests; and Security will use



authentication, role and unauthorized-record-access tests. User ratings will supplement rather than replace technical evidence.

Managers will review bills, correct exceptions and confirm reports; tenants will view bills, report payments and access confirmed receipts. Task completion and difficulties will be recorded. Questionnaire responses will use the approved scale and scoring procedure, while technical results will use counts, percentages, time summaries and failed-case descriptions. Evaluator selection and counts, document counts and questionnaire scoring will be settled with the adviser before final data collection. No completed testing or evaluation result is claimed at the proposal stage.